PROJECT REPORT

FINNEWS AI

AI-Powered Financial News Simplifier for Beginner Investors and Everyday Readers

Submitted for

Capstone Project: Full-Stack AI Web Application

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Project Repository:

https://github.com/prajwalarote05-tech/FinNews-AI

Abstract

FinNews AI is a full-stack web application that fetches real-time financial and business news and turns dense market reports into clear, beginner-friendly explanations. The backend, built with FastAPI, retrieves headlines from NewsAPI, cleans and deduplicates them, and sends each article to Groq's LLM inference service running LLaMA 3.3 70B. The model returns structured JSON containing a simple summary, key takeaways, a "Why It Matters" explanation and a glossary of financial terms. A lightweight dashboard built with HTML, CSS and vanilla JavaScript shows the results as cards, with topic filters, a keyword search box, a details-and-glossary pop-up, a dark/light theme toggle and an About page.

The screenshots in this report were taken with the application running on its built-in sample dataset (articles dated 25-26 September 2026), which the interface announces with a banner. These are sample values for demonstration. The project is an educational tool. It does not provide financial, investment, legal or tax advice, and it never recommends buying, selling or holding any asset.

1. Introduction

Financial news is written mainly for professionals. Headlines and articles are full of acronyms (CPI, FOMC, WTI, H100), accounting terms (gross margin, basis points, yield) and central-bank language that most students, first-time investors and busy readers find hard to follow. As a result, people either skip important news or misread it.

FinNews AI brings together live news retrieval and large-language-model summarisation. It reads each story, keeps the facts that matter (exact figures, dates, percentages and company names) and presents them in plain English, with each difficult term defined alongside. The project sits in the domain of AI for finance and financial literacy: it pairs deterministic data cleaning with AI-generated educational summaries.

2. Problem Statement

Financial news outlets rely heavily on terminology such as *quantitative tightening, basis points, yield inversion, EBITDA compression* and *liquidity crunches*. For college students, beginner investors, non-finance professionals and busy readers this causes several problems:

- Articles are intimidating and time-consuming to read.
- Essential takeaways get hidden behind market jargon.
- Misinterpretation can lead to anxiety or poor financial decisions.
- Many AI tools over-engineer their answers or give risky, unwarranted financial advice.

FinNews AI addresses these problems with a one-click workflow that fetches current stories and explains them safely and simply.

3. Objectives

- Fetch current financial news across stock market, economy, technology, business and global markets topics.
- Clean and deduplicate articles so users see unique, complete stories.

- Simplify every article into a short summary, three key takeaways, a real-world impact note and a glossary.
- Preserve exact figures, dates, percentages and company names, and prevent investment advice, predictions and hallucinated content through strict prompt guardrails.
- Provide a responsive, accessible dashboard with filters, search and a light/dark theme.
- Offer automated tests and a one-click startup script so the project is easy to run and evaluate.

4. Scope and Key Features

4.1 Home Page and Live Statistics

The home page presents the headline "Understand Financial News. Without the Jargon.", a badge stating that the app is powered by Groq Llama 3.3 70B and NewsAPI, and a prominent **Fetch & Simplify** button. A disclaimer banner states that summaries are for informational purposes only and are not financial advice. Four statistic cards summarise the current session: Articles Fetched (7), AI Simplified (6), News Sources (7) and Last Updated (09:30 PM, Oct 30, 2026 in the sample run).

4.2 Search and Topic Filters

A search box accepts keywords such as Inflation, Nvidia, Fed, Oil or S&P 500. Pill-shaped filters let users narrow the feed to **All News, Stock Market, Economy, Technology, Business** or **Global Markets**. A **Refresh News** button in the header reloads the feed.

4.3 News Cards with Simplified Insight

Each article appears as a card showing a topic badge (for example Economy, Technology or Stock Market), a relevant image, the news source (Wall Street Journal, Bloomberg, CNBC and others), the publication date and the headline. Below the headline, a green **Simplified Insight** box starts with "In simple terms:" and gives the plain-English summary. The card then lists key bullet points, a **Why It Matters** callout, and orange glossary chips (for example Interest Rate, Inflation, Federal Reserve, Gross Margin, GPU, S&P 500). Two actions sit at the bottom of each card: **Details & Glossary** and **Read Original Article**.

4.4 Details & Glossary Pop-up

Selecting Details & Glossary opens a modal window with the full simplified view of one article: topic badge, headline, source and date, an **AI Simplified Explanation**, the **Key Takeaways** list, the **Why It Matters** section and the explained financial terms. The window scrolls and can be closed with the X button.

4.5 About Page

The About page, labelled "Project Documentation", explains the problem of financial jargon overload and shows a **System Architecture Flow** diagram. It describes FinNews AI as a decoupled, asynchronous micro-architecture built for speed and reliability. A **Back to News** button returns the user to the feed.

4.6 Theme, Responsiveness and Sample Mode

A theme toggle in the header switches between dark and light modes, and the layout adapts to desktop, tablet and mobile screens. When the NewsAPI and Groq keys are not configured, the app falls back to a real-world financial sample dataset and tells the user so through an

information banner, so the interface can still be demonstrated offline. Adding the keys to the environment file enables live fetching.

5. Technology and Architecture

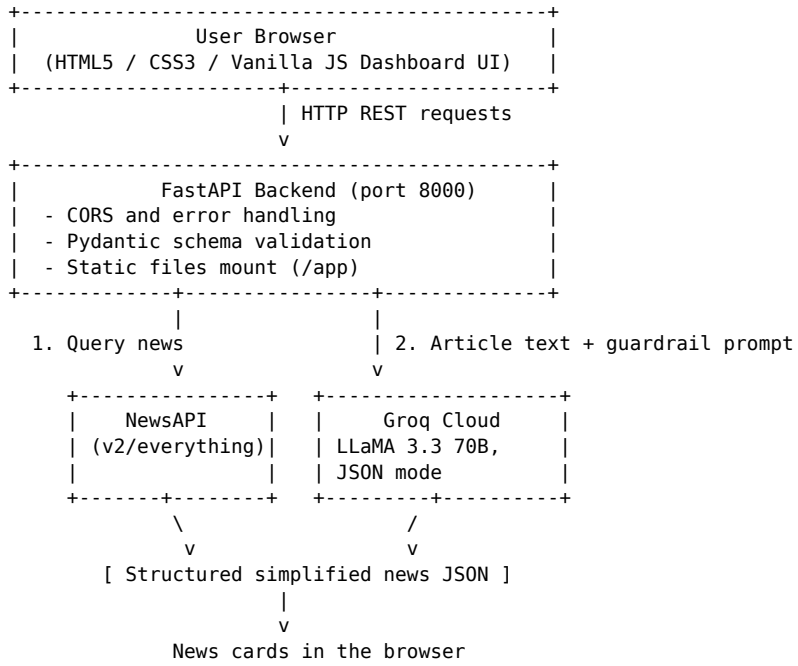
The repository is divided into a *backend*, a *frontend* and a *tests* folder. The stack is summarised below.

Layer	Technology	Purpose
Language	Python 3.12+ (tested on 3.13)	Core backend language
Web framework	FastAPI	High-performance asynchronous API, OpenAPI docs at /docs
Server	Uvicorn (ASGI)	Serves the API and the static frontend
HTTP client	httpx	Async, timeout-managed requests to NewsAPI
Validation	pydantic v2	Request and response schema validation
Configuration	python-dotenv	API keys and model name from environment variables
AI engine	Groq Cloud, LLaMA 3.3 70B Versatile (groq SDK)	Low-latency summarisation with JSON-mode output
News source	NewsAPI	Real-time global financial and business headlines
Frontend	HTML5, CSS3, vanilla JavaScript (ES6+)	Dashboard UI with no build step or heavy framework
Testing	pytest, pytest-asyncio, FastAPI TestClient	Automated tests with mocked external services

5.1 High-Level Workflow

- The user opens the dashboard (served at /app/) and the browser checks /health to confirm the backend is connected.
- The user picks a topic filter and clicks **Fetch & Simplify**. The button is disabled and shows progress: "Fetching latest financial news..." and then "Simplifying with AI...".
- The FastAPI backend queries NewsAPI, then deduplicates the results and removes broken, withdrawn or truncated articles.
- Each cleaned article is sent to Groq with a guardrail system prompt that requests strict JSON.
- The backend validates the JSON with Pydantic and returns it to the browser.
- The frontend renders cards with the summary, key points, Why It Matters callout and glossary chips.

5.2 System Architecture



5.3 Project Structure

```
finnews-ai/
|-- backend/
|   |-- main.py            FastAPI app, endpoints, CORS, static hosting
|   |-- config.py          Environment variable configuration
|   |-- news_service.py    NewsAPI fetcher, query mapper, deduplication
|   |-- ai_service.py      Groq client, prompt, JSON parser/validator
|   |-- schemas.py         Pydantic request/response models
|   |-- requirements.txt   Python dependencies
|   `-- .env.example       Environment template
|-- frontend/
|   |-- index.html         User interface structure
|   |-- style.css          Dark financial-tech styling
|   `-- script.js          State, API integration, card renderer
|-- tests/
|   |-- test_health.py     Health and root endpoint tests
|   `-- test_api.py        News and simplify tests with mocks
|-- .gitignore, README.md, pytest.ini
`-- run.bat                One-click startup script for Windows
```

5.4 API Endpoints

Method	Endpoint	Description	Parameters
GET	/	API status and links	None
GET	/health	Service health status	None
GET	/api/news	Raw, deduplicated news	?topic=all&page_size=5
POST	/api/simplify	Simplify a single article	JSON SimplifyRequest
POST	/api/news/simplify-all	Batch fetch and simplify	?topic=all&limit=3

6. AI Design and Output Format

The AI layer is built for safety and consistency rather than open-ended generation. The system prompt instructs the model to give no investment advice, make no stock predictions, stay

faithful to the supplied article, and keep exact figures, dates, percentages and company names. Groq's JSON mode enforces a fixed structure, which the backend validates before the UI renders it. The model name is read from the GROQ_MODEL environment variable, so it can be changed without editing code.

Field	Meaning
simple_summary	2-3 sentence plain-English explanation of the story.
key_points	Three bullet points that highlight the main facts.
why_it_matters	Real-world impact for everyday consumers and investors.
terms_explained	List of {term, meaning} pairs that define difficult words.
source, published_at, url	Metadata carried through from the original article.

7. Screenshots of the Application

The following figures are screenshots of the running application and document the implemented interface. They were captured on 30 October 2026 using the sample dataset.

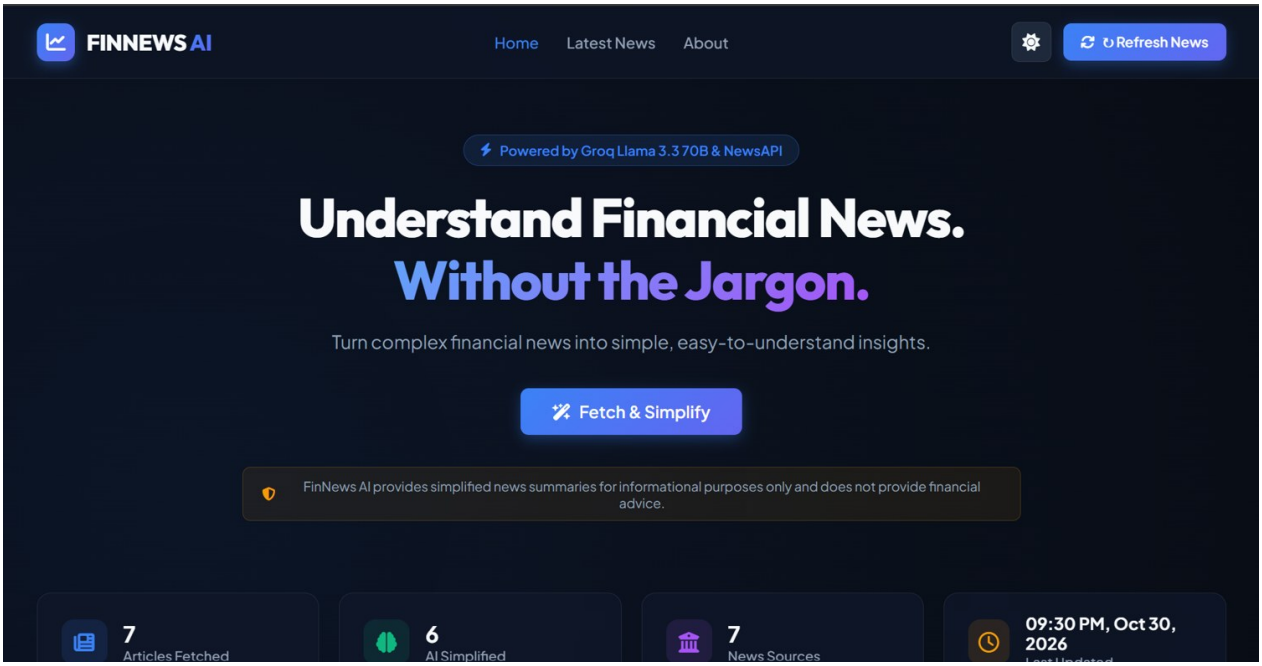


Figure 1. Home page with the Fetch & Simplify button, disclaimer banner and live statistics cards.

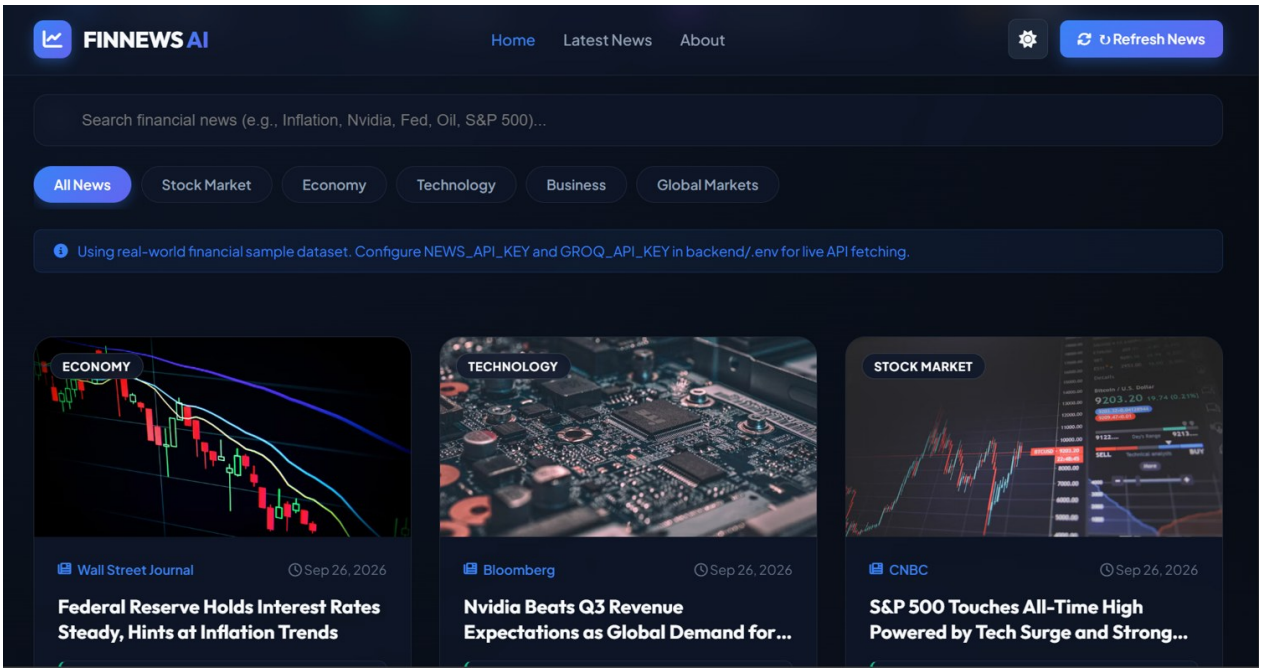


Figure 2. News feed with search box, topic filters, sample-data notice and article cards.

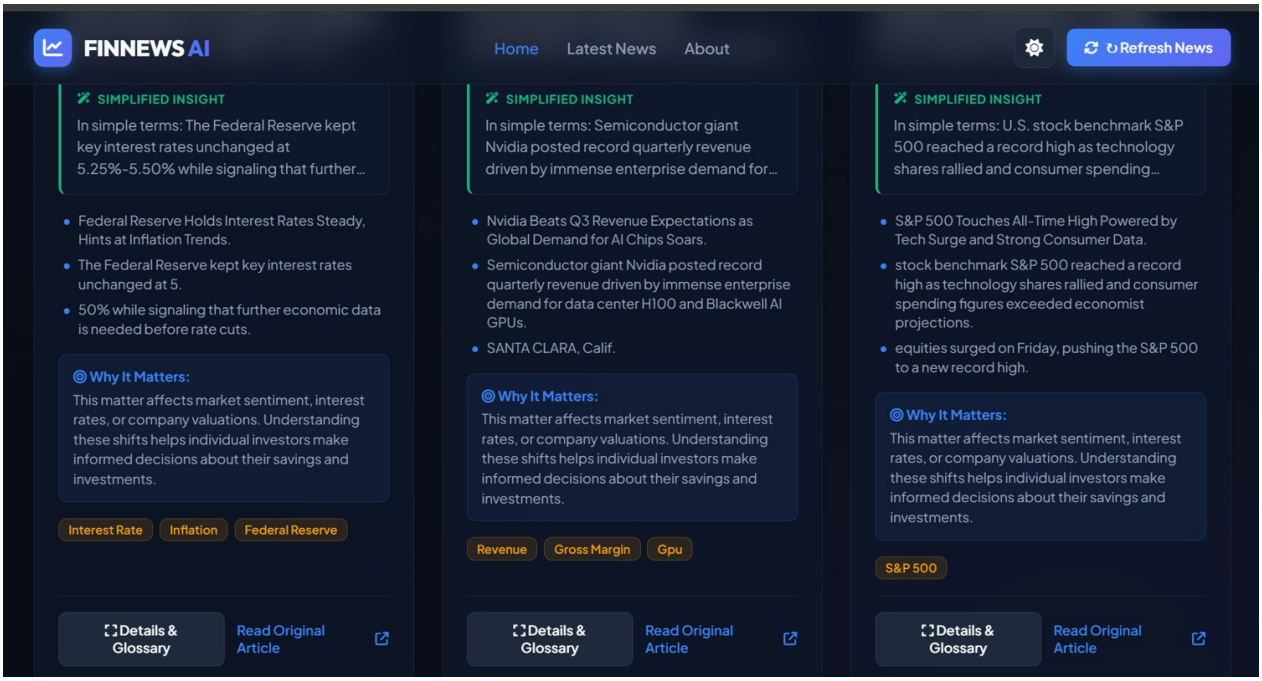


Figure 3. Article cards showing the Simplified Insight, key points, Why It Matters callout and glossary chips.

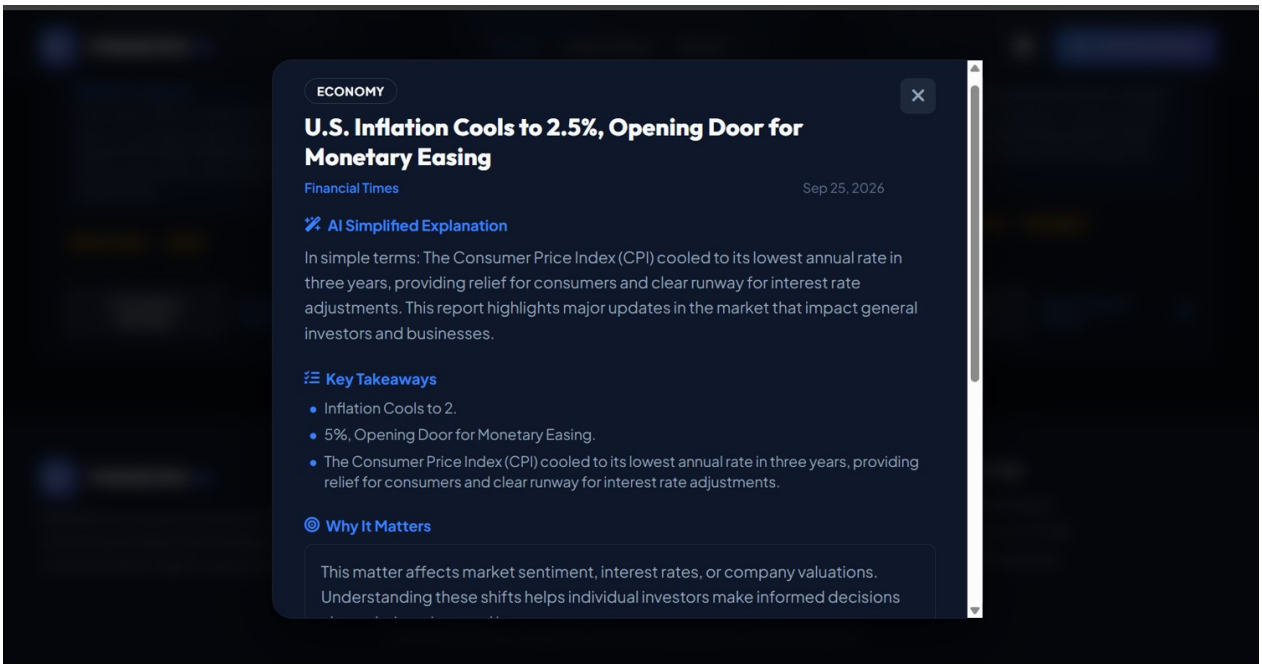


Figure 4. Details & Glossary pop-up with the AI Simplified Explanation and Key Takeaways.

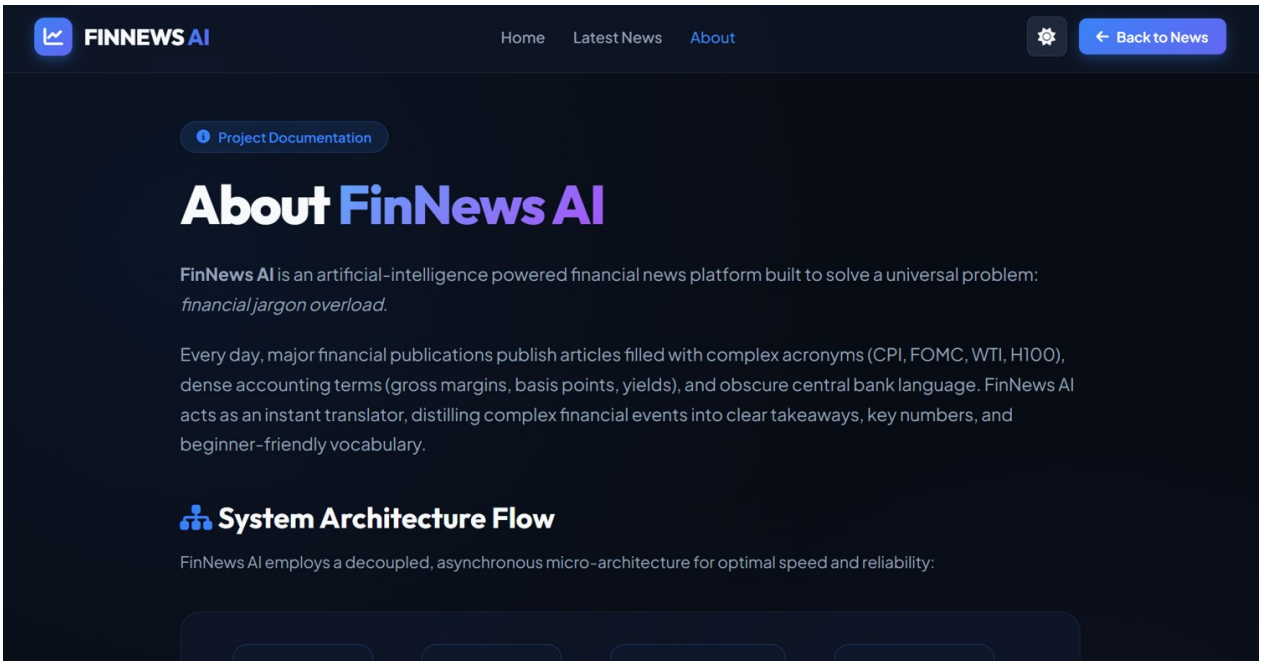


Figure 5. About page describing the project and its system architecture flow.

8. Security, Privacy and Responsible AI

- API keys are loaded from environment variables through python-dotenv. The .env file is excluded from Git by .gitignore and must never be committed; only the placeholder file .env.example is stored in the repository.
- The prompt forbids investment advice, predictions and hallucinations, and the interface and README carry a visible disclaimer that summaries are educational only.
- Requests and responses are validated with Pydantic, and a missing NewsAPI key returns a controlled error instead of a crash.

- Only article text (title, description and content) is sent to the AI provider. No personal user data is collected or stored.
- AI output should be treated as untrusted text. JSON mode and schema validation reduce malformed responses, and the sample-dataset fallback keeps the app usable when an external service fails.
- For any public deployment, serve the app over HTTPS and restrict the allowed CORS origins.

9. Testing and Evaluation

The tests directory contains an automated suite built with pytest, pytest-asyncio and the FastAPI TestClient. External network calls are mocked, so the tests run quickly and do not use live API tokens. The suite is run with `pytest -v` and the README reports 7 passing tests.

Test	What it checks
test_root_endpoint	Root endpoint returns API status.
test_health_endpoint	Health check endpoint responds correctly.
test_simplify_validation_missing_title	Validation rejects a request without a title.
test_simplify_success	A single article is simplified correctly (mocked AI).
test_get_news_missing_api_key	Clean error when the NewsAPI key is missing.
test_get_news_success	News retrieval and deduplication (mocked NewsAPI).
test_simplify_all_news_success	Batch fetch-and-simplify flow.

Manual checks on the running interface covered the home page, topic filters, card rendering, the Details & Glossary pop-up, the About page and the theme toggle, as shown in the screenshots.

10. Installation and Running

- Install Python 3.12 or later and open the project folder (`cd finnews-ai`).
- Create and activate a virtual environment (`python -m venv venv`, then `venv\Scripts\activate` on Windows).
- Install dependencies with `pip install -r backend/requirements.txt`.
- Create a .env file from .env.example and add NEWS_API_KEY, GROQ_API_KEY and GROQ_MODEL (default llama-3.3-70b-versatile). Both keys are free: NewsAPI from newsapi.org and Groq from console.groq.com.
- On Windows, double-click run.bat for one-click setup. Otherwise run `uvicorn backend.main:app --reload --host 127.0.0.1 --port 8000`.
- Open http://127.0.0.1:8000/app/ for the dashboard, /docs for the Swagger UI and /health for the health check.

11. Limitations

- The free NewsAPI tier allows about 100 requests per day and delays articles by up to 24 hours.

- Free NewsAPI responses truncate article bodies at about 200 characters, so the app combines title and description when the body is cut off.
- The free Groq tier has per-minute limits, so batch requests are capped at 3-5 articles.
- In sample-dataset mode, the simple rule-based summary can split sentences at decimal points (for example "5.25%" appearing as "5." and "25%" on separate lines in Figures 3 and 4). Live AI mode produces proper summaries.
- AI summaries can be incomplete or inaccurate and should be checked against the original article, which every card links to.

12. Future Enhancements

- Audio briefings using browser SpeechSynthesis or an AI voice model.
- Search by company ticker (for example $AAPL, $TSLA) in addition to keywords.
- Bookmarks and saved cards stored in browser localStorage.
- Multi-language summaries in Hindi, Spanish, French and Mandarin.
- A sentiment gauge showing positive, neutral or cautious tone for each article.
- Evaluation of AI summaries against a test set for factuality, consistency and safety.

13. Conclusion

FinNews AI shows how a modern LLM can be combined with a real-time data source to make financial information accessible to everyone. A fast FastAPI backend, NewsAPI curation, Groq's low-latency LLaMA 3.3 70B inference and strict safety guardrails turn jargon-heavy reporting into short, understandable summaries. The simple setup, automated tests and dependency-free frontend make the project easy to run, evaluate and extend. The next steps are to run it against live data at scale, fix the sentence splitting in sample mode, and evaluate the AI summaries more formally.

Disclaimer: FinNews AI provides simplified summaries for educational and informational purposes only. It does not provide financial, investment, legal or tax advice, and it does not recommend buying, selling or holding any security, stock, cryptocurrency or financial instrument. Always consult a certified financial advisor before making investment decisions.

References

- Project repository: https://github.com/prajwalarote05-tech/FinNews-AI
- FastAPI documentation: https://fastapi.tiangolo.com/
- Groq documentation: https://console.groq.com/docs
- NewsAPI documentation: https://newsapi.org/docs
- Pydantic documentation: https://docs.pydantic.dev/
- pytest documentation: https://docs.pytest.org/